

WHY $\text{Comm}_N \leq CF_N$ IS CRUDE, AND WHAT TO REPLACE IT WITH

Companion to “Fluctuations around the mean-field limit for Coulomb/Riesz flows” [N]. Labels: **Fact** = proved; **Hope** = target statement with its hypotheses; **Missing ingredient** = what it rests on; **Obstruction** = what these ideas do not reach. Revised twice. The first version claimed the additive constant $\mathfrak{o}_N$ was the whole loss; it is not, the multiplicative bound itself is crude. The second invoked $\text{div } u = 0$, which holds only in the Hamiltonian logarithmic case; [Fact 3.2](#) below replaces that accident by an identity valid for every mean-field velocity.

1. SETTING, AND THE ONE IDENTITY EVERYTHING RUNS THROUGH

1.1. Setting. $\mathbb{D} = \mathbb{T}^d$ or $\mathbb{R}^d$; $\mathbf{g}(x) = |x|^{-s}$, $0 < s < d$ ($s = 0$: $-\log|x|$, mentioned where it differs). $\mathbb{M}$ is a constant matrix with $\mathbb{S} := \text{sym } \mathbb{M} \leq 0$; $\mathbb{M} = -\text{Id}$ is the gradient flow, $\mathbb{M} = J$ (rotation) the 2D Hamiltonian case. Particles solve

$$dx_i = \left(\frac{1}{N} \sum_{j \neq i} \mathbb{M} \nabla \mathbf{g}(x_i - x_j) - \nabla V(x_i) \right) dt + \sqrt{2/\beta} dW_i,$$

$\mu_N := \frac{1}{N} \sum \delta_{x_i}$, and μ^t solves the mean-field equation $\partial_t \mu = -\text{div}(\mu u) + \beta^{-1} \Delta \mu$, $u := \mathbb{M} \nabla \mathbf{g} * \mu - \nabla V$. Set $\rho_N := \mu_N - \mu$,

$$a_i := \frac{1}{N} \sum_{j \neq i} \nabla \mathbf{g}(x_i - x_j) - \nabla \mathbf{g} * \mu(x_i) = \int_{y \neq x_i} \nabla \mathbf{g}(x_i - y) d\rho_N(y), \quad D_N := \frac{1}{N} \sum_i |a_i|^2,$$

so that $dx_i = (u(x_i) + \mathbb{M} a_i) dt + \sqrt{2/\beta} dW_i$. The modulated energy and the commutator are

$$F_N := \frac{1}{2} \iint_{x \neq y} \mathbf{g}(x-y) d\rho_N^{\otimes 2}, \quad \text{Comm}_N(v) := \iint_{x \neq y} K_v d\rho_N^{\otimes 2}, \quad K_v(x, y) := (v(x) - v(y)) \cdot \nabla \mathbf{g}(x-y).$$

For Φ symmetric on $\mathbb{D}^2$, $\langle \Phi, \rho_N^{\otimes 2} \rangle_{\neq} := \iint_{x \neq y} \Phi d\rho_N^{\otimes 2}$ and $\Phi_\mu(x) := \int \Phi(x, y) d\mu(y)$.

Imported from [N] 1.1. (I1) $F_N(X, \mu) \geq -\mathfrak{o}_N$ for all X , with $\mathfrak{o}_N \asymp N^{s/d-1}$ (plus $\log N/N$ if $s = 0$). (I2) The pathwise commutator inequality $|\text{Comm}_N(v)| \leq C \|v\|_*(F_N + \mathfrak{o}_N)$. (I3) Well-posedness and regularity of the backward linearized equation (3) below, giving the propagator $\mathbf{U}^{r,t}$. (I4) The CLT for the martingale part. (I5) The scaling $\sigma_N = \min(\sqrt{N}\beta, \kappa_N)$ with κ_N the size of the initial fluctuation, the exponent γ of the $H^{-\gamma}$ field theorem, and the master rate. Only (I2) is questioned here.

1.2. The corrector identity. For a deterministic symmetric $\Phi^r(x, y)$, $C^{1,2}$ off the diagonal, define

$$\mathcal{P}_N[\Phi] := \frac{1}{2} \langle \Phi, \rho_N^{\otimes 2} \rangle_{\neq}, \quad \mathcal{A}_r \Phi := \partial_r \Phi + u^r(x) \cdot \nabla_x \Phi + u^r(y) \cdot \nabla_y \Phi + \beta^{-1} (\Delta_x + \Delta_y) \Phi,$$

$$b_i[\Phi] := \int_{y \neq x_i} \nabla_x \Phi(x_i, y) d\rho_N(y), \quad \mathcal{T}_N[\Phi] := \frac{1}{N} \sum_i (\mathbb{M} a_i) \cdot b_i[\Phi].$$

Fact 1.2 (corrector identity).

$$(1) \quad d\mathcal{P}_N[\Phi^r](X^r, \mu^r) = \frac{1}{2} \langle \mathcal{A}_r \Phi^r, \rho_N^{\otimes 2} \rangle_{\neq} dr + \mathcal{T}_N[\Phi^r] dr + \sqrt{2/\beta} \frac{1}{N} \sum_i b_i[\Phi^r] \cdot dW_i^r.$$

Proof. Expand $\mathcal{P}_N[\Phi] = \frac{1}{2N^2} \sum_{i \neq j} \Phi(x_i, x_j) - \frac{1}{N} \sum_i \Phi_\mu(x_i) + \frac{1}{2} \langle \Phi, \mu \otimes \mu \rangle$. Itô on the first sum, with independent Brownian motions and $\nabla_y \Phi(x_i, x_j) = \nabla_x \Phi(x_j, x_i)$, gives

$$\frac{1}{2N^2} \sum_{i \neq j} \mathcal{A}\Phi(x_i, x_j) dr + \frac{1}{N^2} \sum_{i \neq j} \nabla_x \Phi(x_i, x_j) \cdot \mathbb{M}a_i dr + \frac{1}{N^2} \sum_i \left(\sum_{j \neq i} \nabla_x \Phi(x_i, x_j) \right) \cdot \sqrt{2/\beta} dW_i.$$

For the second sum, $\partial_r \Phi_\mu = (\partial_r \Phi)_\mu + \int \Phi \partial_r \mu$ and, integrating the mean-field equation by parts, $\int \Phi \partial_r \mu dy = \int [u(y) \cdot \nabla_y \Phi + \beta^{-1} \Delta_y \Phi] \mu dy$; hence $d(-\frac{1}{N} \sum_i \Phi_\mu(x_i)) = -\frac{1}{N} \sum_i (\mathcal{A}\Phi)_\mu(x_i) dr - \frac{1}{N} \sum_i \nabla \Phi_\mu(x_i) \cdot \mathbb{M}a_i dr - \frac{1}{N} \sum_i \nabla \Phi_\mu(x_i) \cdot \sqrt{2/\beta} dW_i$. The third term gives $\frac{1}{2} \langle \mathcal{A}\Phi, \mu \otimes \mu \rangle dr$. Summing, the $\mathcal{A}\Phi$ terms assemble into $\frac{1}{2} \langle \mathcal{A}\Phi, \rho_N^{\otimes 2} \rangle_\neq$ and the $\mathbb{M}a_i$ terms into $\frac{1}{N} \sum_i \mathbb{M}a_i \cdot [\frac{1}{N} \sum_{j \neq i} \nabla_x \Phi(x_i, x_j) - \nabla \Phi_\mu(x_i)] = \mathcal{T}_N[\Phi]$. $\square$

Fact 1.3 (energy identity). *Taking $\Phi = \mathbf{g}$: $\mathcal{A}\mathbf{g} = K_u + 2\beta^{-1} \Delta \mathbf{g}$ and $b_i[\mathbf{g}] = a_i$, so*

$$(2) \quad dF_N = \left[\frac{1}{2} \text{Comm}_N(u) + \frac{1}{N} \sum_i a_i^T \mathbb{S}a_i + \beta^{-1} \langle \Delta \mathbf{g}, \rho_N^{\otimes 2} \rangle_\neq \right] dr + \sqrt{2/\beta} \frac{1}{N} \sum_i a_i \cdot dW_i.$$

For the gradient flow the middle term is $-D_N$.

Fact 1.4 (duality identity). *Let $\psi_f(y) := \int \nabla f(x) \cdot \mathbb{M} \nabla \mathbf{g}(x - y) d\mu(x)$ (smooth) and $\mathcal{L}_\mu f := u \cdot \nabla f + \beta^{-1} \Delta f + \psi_f$. Then*

$$d\langle f^r, \rho_N^r \rangle = \langle \partial_r f + \mathcal{L}_{\mu^r} f, \rho_N^r \rangle dr + \frac{1}{2} \text{Comm}_N(\mathbb{M}^* \nabla f^r) dr + \sqrt{2/\beta} \frac{1}{N} \sum_i \nabla f(x_i) \cdot dW_i.$$

Hence if $f^r = U^{r,t} \phi$ solves the backward linearized equation

$$(3) \quad \partial_r f + \mathcal{L}_{\mu^r} f = 0, \quad f^t = \phi,$$

then, with $M_N^t[f]$ the martingale,

$$(4) \quad \langle \phi, \rho_N^t \rangle = \langle U^{0,t} \phi, \rho_N^0 \rangle + M_N^t[f] + \frac{1}{2} \int_0^t \text{Comm}_N(\mathbb{M}^* \nabla f^r) dr.$$

Proof. $d\langle f, \mu_N \rangle = \frac{1}{N} \sum_i [\partial_r f + (u + \mathbb{M}a_i) \cdot \nabla f + \beta^{-1} \Delta f](x_i) dr + \text{noise}$, and $d\langle f, \mu \rangle = \langle \partial_r f + u \cdot \nabla f + \beta^{-1} \Delta f, \mu \rangle dr$. The force term is $\frac{1}{N} \sum_i \mathbb{M}a_i \cdot \nabla f(x_i) = \int \int_\neq \nabla f(x) \cdot \mathbb{M} \nabla \mathbf{g}(x - y) d\mu_N(x) d\rho_N(y)$; writing $\mu_N = \mu + \rho_N$ in the first slot gives $\langle \psi_f, \rho_N \rangle + \int \int_\neq \nabla f(x) \cdot \mathbb{M} \nabla \mathbf{g}(x - y) d\rho_N^{\otimes 2}$, and the last integral symmetrizes to $\frac{1}{2} \text{Comm}_N(\mathbb{M}^* \nabla f)$ since $(\nabla f(x) - \nabla f(y)) \cdot \mathbb{M} \nabla \mathbf{g} = (\mathbb{M}^*(\nabla f(x) - \nabla f(y))) \cdot \nabla \mathbf{g}$. $\square$

Multiplying (4) by σ_N gives the identity of [N]. The first two terms yield the limit law (I3, I4); the theorem holds as soon as the last term is $o(\sigma_N^{-1})$. Everything below is about it. [N] estimates it by (I2).

2. TWO EXACT REWRITINGS

Fact 2.1 (force form). *Exactly, with $h_v := \nabla \mathbf{g} * (v\mu)$ (a Riesz potential of a smooth function, hence smooth):*

$$(5) \quad \text{Comm}_N(v) = \frac{2}{N} \sum_i v(x_i) \cdot a_i + 2\langle h_v, \rho_N \rangle.$$

Proof. $\text{Comm}_N(v) = \frac{1}{N^2} \sum_{i \neq j} (v(x_i) - v(x_j)) \cdot \nabla \mathbf{g}(x_i - x_j) - 2\langle K_{v,\mu}, \mu_N \rangle + \langle K_{v,\mu}, \mu \rangle$. By symmetry the double sum is $\frac{2}{N^2} \sum_{i \neq j} v(x_i) \cdot \nabla \mathbf{g}(x_i - x_j) = \frac{2}{N} \sum_i v(x_i) \cdot a_i + \frac{2}{N} \sum_i v(x_i) \cdot \nabla \mathbf{g} * \mu(x_i)$. Since $K_{v,\mu}(x) = v(x) \cdot \nabla \mathbf{g} * \mu(x) - h_v(x)$, the last two groups combine to $2\langle h_v, \mu_N \rangle + \langle K_{v,\mu}, \mu \rangle$, and $\langle h_v, \mu \rangle = -\langle v \cdot \nabla \mathbf{g} * \mu, \mu \rangle = -\langle K_{v,\mu}, \mu \rangle - \langle h_v, \mu \rangle$ forces $\langle K_{v,\mu}, \mu \rangle = -2\langle h_v, \mu \rangle$, whence $2\langle h_v, \mu_N \rangle + \langle K_{v,\mu}, \mu \rangle = 2\langle h_v, \rho_N \rangle$. $\square$

(Verified numerically on $\mathbb{T}^1$, $s = 1/2$, $\mu \equiv 1$, $v = \sin 2\pi x$, $N = 200, 400, 800$: both sides agree to 10^{-11} ; script attached.) The singular kernel appears nowhere except inside a_i , which the equation produces anyway.

Fact 2.2 (stress form). *With $m = \frac{x+y}{2}$, $z = x - y$, $E_v = \text{sym } \nabla v$, E_v° its traceless part, define the local energy and stress measures*

$$e_N := \frac{1}{2} \iint_{\neq} \mathbf{g}(z) \delta_m d\rho_N^{\otimes 2}, \quad \mathcal{S}_N^\circ := -s \iint_{\neq} |z|^{-s-2} (z \otimes z)^\circ \delta_m d\rho_N^{\otimes 2}.$$

Then $e_N(\mathbb{D}) = F_N$ and

$$(6) \quad \text{Comm}_N(v) = -\frac{2s}{d} \langle \text{div } v, e_N \rangle + \langle E_v^\circ, \mathcal{S}_N^\circ \rangle + \langle r_v, \rho_N^{\otimes 2} \rangle_{\neq}, \quad |r_v| \leq C \|\nabla^2 v\|_\infty |z|^{1-s},$$

with, exactly and with no remainder, $\text{Comm}_N(a\text{Id}) = -2saF_N$; and in $d = 1$, $\text{Comm}_N(v) = -2sF_N^{w_v}$, $w_v(x, y) = \frac{v(x)-v(y)}{x-y}$ smooth.

Proof. With m, z as above, $v(x) - v(y) = v(m + \frac{z}{2}) - v(m - \frac{z}{2}) = \nabla v(m)z + O(\|\nabla^2 v\|_\infty |z|^2)$, and $\nabla \mathbf{g}(z) = -s|z|^{-s-2}z$, so $K_v = -s|z|^{-s-2}z \cdot \nabla v(m)z + O(\|\nabla^2 v\|_\infty |z|^{1-s})$. Only $E_v = \text{sym } \nabla v$ survives in $z \cdot \nabla v z$, and $z \cdot E_v z = \frac{\text{tr } E_v}{d} |z|^2 + z \cdot E_v^\circ z$ with $\text{tr } E_v = \text{div } v$. Integrating against $\rho_N^{\otimes 2}$ gives (6). For $v = ax$ the expansion is exact with $E_v^\circ = 0$, giving $\text{Comm}_N(a\text{Id}) = -2saF_N$. In $d = 1$, $(v(x) - v(y))g'(z) = w_v(x, y)zg'(z) = -sw_v \mathbf{g}(z)$ exactly. (For $s = 0$: $z \cdot \nabla \mathbf{g} = -1$, so the compression term is $-\frac{1}{d} \langle \text{div } v(m), \rho_N^{\otimes 2} \rangle_{\neq}$ and $\text{Comm}_N(a\text{Id}) = a/N$.) $\square$

Equations (5) and (6) are the same identity before and after discrete integration by parts: $\text{div } \mathcal{S}_N$ is the interparticle force density (Irving–Kirkwood).

3. THREE WAYS THE BOUND IS CRUDE

(C1) **It reads a transport identity as an inequality.** The compression term in (6) is $-\frac{2s}{d} \langle \text{div } u, e_N \rangle$, and in local equilibrium $e_N \approx \bar{e}_N$, where $\bar{e}_N := \varepsilon_s N^{s/d-1} (\mu^t)^{1+s/d} dx$. Along the mean-field flow $\frac{d}{dt} \int \mu^{1+s/d} = -\frac{s}{d} \int \mu^{1+s/d} \text{div } u$, so

$$(7) \quad -\frac{s}{d} \langle \text{div } u, \bar{e}_N \rangle = \frac{d}{dt} \left(\varepsilon_s N^{s/d-1} \int (\mu^t)^{1+s/d} \right) = \frac{d}{dt} \bar{F}_N,$$

exactly, with the correct exponent, and for every sign of $\text{div } u$. The compression term is therefore not an error at all: it is the transport of the self-energy density, which the mean-field flow dilutes or concentrates. The bound $|\text{Comm}_N(u)| \leq C \|\nabla u\|_\infty F_N$ converts this identity into growth e^{Ct} . (Only in the incompressible case $\text{div } u \equiv 0$, e.g. the Hamiltonian 2D log case $\mathbb{M} = J$, is $\bar{F}_N$ constant and the term absent outright; that is a special case of (7), not the general mechanism.)

(C2) **It replaces a measure by its total mass.** $\text{Comm}_N(v)$ pairs ∇v against e_N and $\mathcal{S}_N^\circ$ pointwise; the bound uses only $e_N(\mathbb{D}) = F_N$. Any spatial correlation between where energy sits and where the strain is — e.g. $\text{div } u$ small on the bulk of the support — is discarded. Equivalently, in the force form (5): the estimate bounds a distribution by its mass instead of integrating by parts and using that $\mathcal{S}_N$ is nearly momentum-balanced.

(C3) **It treats the shear term as of the size of the energy.** $\langle E_v^\circ, \mathcal{S}_N^\circ \rangle$ is bounded by $\|\nabla v\|_\infty F_N$, but $\mathcal{S}_N^\circ$ vanishes for any locally isotropic structure, and in Regime P it is of order (relaxation time) $\times$ (strain rate) $\times F_N \ll F_N$.

(C4) **Only then:** the additive $\mathbf{o}_N$, which prevents F_N from being used as a small quantity even when the configuration is perfectly ordered.

Remark 3.1 (what (5) does and does not buy). It is tempting to read off $|\text{Comm}_N(v)| \leq 2\|v\|_{L^2(\mu_N)} D_N^{1/2} + 2\|h_v\|_{C^\gamma} \|\rho_N\|_{H^{-\gamma}}$ and conclude that the dissipation replaces the energy. Two

caveats, both real. (i) The two terms of (5) each contain a linear-in- ρ_N piece of size $\|\rho_N\|$, and these cancel: to first order $\frac{2}{N} \sum v(x_i) \cdot a_i \approx -2\langle h_v, \rho_N \rangle$. Bounding the two pieces separately is therefore *worse* than the pathwise bound, not better — each piece alone is of the size of the answer. (ii) Antisymmetrizing the near-diagonal part of $\frac{2}{N} \sum v \cdot a_i$ returns $\langle \nabla v, \mathcal{S}_N^{\text{near}} \rangle$, i.e. the energy bound again. So (5) is not by itself an improvement; its value is diagnostic ((C2): all the cancellation lives in $\text{div } \mathcal{S}_N$) and it is the right form for the *mesoscopic* part of the force, where h_v and the far field combine into a smooth quadratic form of ρ_N with no singular kernel at all.

Conclusion of the diagnosis. (6) is an *equality*; the pathwise bound is that equality with absolute values. The fix is not a better inequality but to carry e_N and $\mathcal{S}_N^\circ$ as objects, and to subtract from F_N the part of it that the mean-field flow merely transports.

Fact 3.2 (renormalized energy: no forcing, no floor). *Let ε_s be the energy per unit volume of the local structure at unit density (in $d = 1$ at $\beta = \infty$, $\varepsilon_s = \zeta(s)$, the lattice value); by scaling, the local energy density at density ρ is $\varepsilon_s N^{s/d-1} \rho^{1+s/d}$, which is the meaning of $\bar{e}_N$ above. Set*

$$G_N := F_N - \bar{F}_N, \quad \bar{F}_N := \varepsilon_s N^{s/d-1} \int (\mu^t)^{1+s/d} dx.$$

Then, for any mean-field velocity u ,

$$(8) \quad \frac{d}{dt} G_N = \underbrace{-\frac{s}{d} \langle \text{div } u, e_N - \bar{e}_N \rangle}_{\text{fluctuation of the local energy}} + \underbrace{\frac{1}{2} \langle E_u^\circ, \mathcal{S}_N^\circ \rangle}_{\text{shear}} + \frac{1}{2} \langle r_u, \rho_N^{\otimes 2} \rangle_\neq + \frac{1}{N} \sum_i a_i^T \mathbb{S} a_i + \beta^{-1}(\dots).$$

No compression forcing survives, and G_N carries no self-energy floor: the $N^{s/d-1}$ that F_N can never go below has been subtracted, not estimated.

Proof. Insert (6) with $v = u$ into (2), write $e_N = \bar{e}_N + (e_N - \bar{e}_N)$ in the compression term, and use (7): $\frac{d}{dt} \int (\mu^t)^{1+s/d} = (1 + \frac{s}{d}) \int \mu^{s/d} \partial_t \mu = -(1 + \frac{s}{d}) \int \mu^{s/d} \text{div}(\mu u) = -\frac{s}{d} \int \mu^{1+s/d} \text{div } u$ at $\beta = \infty$ (the diffusive part of $\partial_t \mu$ contributes $\beta^{-1}(1 + \frac{s}{d}) \int \mu^{s/d} \Delta \mu$, absorbed into the $\beta^{-1}(\dots)$). $\square$

This is the general- d , general- u form of the endpoint cancellation (12). In the duality identity the same split sends $\bar{e}_N$ into the explicit pressure correction ν^t of Hope 5.1 — a term to be *computed*, not an error — and leaves only $e_N - \bar{e}_N$ to estimate. Read statically, (7) is the $\ell = 0$ (charge) sum rule of the free-energy expansion, and ν^t is the dynamic form of the β -independent scaling part ($-\frac{1}{4}$ in the 2D log case) of the next-order mean: the compression side is finished in both programs, and whatever effort remains belongs to shear.

4. THE REPLACEMENT: A WEIGHTED ENERGY HIERARCHY

For symmetric weights $w(x, y)$ set $F_N^w := \frac{1}{2} \int \int_\neq w g d\rho_N^{\otimes 2}$ (so $F_N^1 = F_N$, and $F_N^w \approx \langle w(x, x), e_N \rangle$). These are the localized energies (C2) asks for.

Fact 4.1 ($d = 1$: the hierarchy closes exactly). *In $d = 1$, with $\mathbb{M} = \pm 1$, $\tilde{w} := \frac{\partial_x w - \partial_y w}{x - y}$ (smooth, since w is symmetric), $D_N^w := \frac{1}{N} \sum_i w(x_i, x_i) |a_i|^2$:*

$$(9) \quad \text{Comm}_N(v) = -2s F_N^{w_v},$$

$$(10) \quad dF_N^{w^r} = \left[F_N^{Aw} - s F_N^{w w_u} - 2s \beta^{-1} F_N^{\tilde{w}} + \beta^{-1} \langle w g'', \rho_N^{\otimes 2} \rangle_\neq + \mathbb{M} D_N^w + \mathcal{R}_N^w \right] dr + d(\text{mart.}),$$

where $\mathcal{R}_N^w := \mathbb{M} \frac{1}{N} \sum_i a_i \left[\int_\neq (w(x_i, y) - w(x_i, x_i)) g'(x_i - y) d\rho_N(y) + \int_\neq \partial_x w(x_i, y) g(x_i - y) d\rho_N(y) \right]$ carries only kernels less singular than g . So $\{F_N^w\}$ is closed modulo the weighted dissipation and less singular remainders — no inequality, and the sign of $\mathbb{M} D_N^w$ is definite for the gradient flow.

Proof. (9) is Fact 2.2. For (10) apply (1) to $\Phi = wg$. Using $\partial_x g(x-y) = g'(z) = -\partial_y g(x-y)$ and $zg'(z) = -sg(z)$: $(u(x)\partial_x + u(y)\partial_y)(wg) = g(u(x)\partial_x w + u(y)\partial_y w) - swu_g$ and $(\partial_x^2 + \partial_y^2)(wg) = g(\partial_x^2 + \partial_y^2)w + 2g'(\partial_x w - \partial_y w) + 2wg'' = g(\partial_x^2 + \partial_y^2)w - 2s\tilde{w}g + 2wg''$, which gives (??) and hence the first four terms of (10). For $\mathcal{T}_N[wg]$: $b_i[wg] = \int_{\neq} [\partial_x w g + wg'] (x_i, y) d\rho_N(y)$, and $\int_{\neq} w(x_i, y) g'(x_i - y) d\rho_N(y) = w(x_i, x_i) a_i + \int_{\neq} (w(x_i, y) - w(x_i, x_i)) g' d\rho_N$; multiplying by $\mathbb{M} a_i$ and averaging gives $\mathbb{M} D_N^w + \mathcal{R}_N^w$. $\square$

Two consequences.

(a) *The remainder becomes an explicit smooth corrector.* At $\beta = \infty$ let W^r solve on $\mathbb{T}^1 \times \mathbb{T}^1$ the linear transport equation $\partial_r W + u(x)\partial_x W + u(y)\partial_y W - sw_u W = sw_{v^r}$, $W^t = 0$ ($v^r = \mathbb{M}\partial_x f^r$). By (??) at $\beta = \infty$, $\mathcal{A}(Wg) = sw_v g = -K_v$, so integrating (1) on $[0, t]$ gives $\frac{1}{2} \int_0^t \text{Comm}_N(v^r) dr = F_N^{W^0}(X^0, \mu^0) + \int_0^t \mathcal{T}_N[W^r g] dr$, and by (??) with $\mathbb{M} = -1$, (4) becomes exact:

$$(11) \quad \langle \phi, \rho_N^t \rangle = \langle U^{0,t} \phi, \rho_N^0 \rangle + F_N^{W^0}(X^0, \mu^0) - \int_0^t D_N^{W^r} dr + \int_0^t \mathcal{R}_N^{W^r} dr + \text{mart.}$$

(b) *The self-energy floor cancels between endpoints.* Let Ψ^t be the mean-field flow map ($\partial_t \Psi^t(y) = u^t(\Psi^t y)$), so $\mu^t = \Psi_{\#}^t \mu^0$ and, for $w_0 \geq 0$, let w^r solve the homogeneous equation $\partial_r w + u(x)\partial_x w + u(y)\partial_y w - sw_u w = 0$, $w^t = w_0$; then $w^r > 0$. On the diagonal $w_u(x, x) = u'(x)$, so along $r \mapsto (\Psi^r y, \Psi^r y)$ one has $\frac{d}{dr} w^r(\Psi^r y, \Psi^r y) = s u'(\Psi^r y) w^r$, and since $\frac{d}{dr} \log \partial_y \Psi^r(y) = u'(\Psi^r y)$,

$$w^0(y, y) = w_0(\Psi^t y) (\partial_y \Psi^t(y))^{-s}, \quad \text{so that} \quad \int w_0(x) (\mu^t)^{1+s}(x) dx = \int w^0(y, y) (\mu^0)^{1+s}(y) dy$$

by the change of variables $x = \Psi^t y$, $\mu^t(\Psi^t y) = \mu^0(y)/\partial_y \Psi^t(y)$: the transported weight matches exactly the transport of μ^{1+s} . At $\beta = \infty$, $\mathbb{M} = -1$, (10) with this w^r integrates to

$$(12) \quad \int_0^t D_N^{w^r} dr = F_N^{w_0}(X^0, \mu^0) - F_N^{w_0}(X^t, \mu^t) + \int_0^t \mathcal{R}_N^{w^r} dr.$$

Each F_N separately sits at the crystalline floor $\asymp N^{s-1}$; the *difference* does not, provided a matching lower bound holds at both times. This is the precise sense in which (C4) is dissolved rather than estimated.

Fact 4.2 ($d \geq 2$: what closes). *The scalar-weighted hierarchy closes iff $E_u^\circ \equiv 0$, i.e. for dilational flows in any d and for conformal flows in the 2D logarithmic case. Otherwise compression couples to S_N° , S_N° to the next multipole, and so on.*

Why. Repeating the computation of Fact 4.1 in $d \geq 2$, $\mathcal{A}(wg) = (\mathcal{A}w)g + wK_u + 2\beta^{-1}(\nabla_x w - \nabla_y w) \cdot \nabla g + 2\beta^{-1}w\Delta g$, and by (6) $wK_u = -\frac{s}{d}w \operatorname{div} u g - sw|z|^{-s-2}z \cdot E_u^\circ z + wr_u$. The middle term is not of the form (smooth weight) $\times g$ unless $E_u^\circ = 0$. $\square$

5. WHAT WE HOPE TO PROVE

Hope 5.1 ($d = 1$, $\beta = \infty$, gradient flow, lattice-like data). $0 < s < 1$, X^0 a perturbed μ^0 -quantile lattice, $\kappa_N \ll N^{1-s+\delta}$. *Then*

$$\mu_N^t = \mu^t + N^{s-1}\nu^t + o(N^{s-1} \wedge \kappa_N^{-1}), \quad \partial_t \nu = \mathcal{L}_\mu^* \nu + s \Delta(\zeta(s)\mu^{1+s}), \quad \nu^0 = 0,$$

and $\kappa_N(\rho_N^t - N^{s-1}\nu^t)$ is transported by U as in [N]. *The remainder is not discarded: it is identified as the pressure of the correlation hole.*

Proof route: (11) with $\int D_N^W$ controlled by (12). Rests on Missing ingredient 6.1 and Missing ingredient 6.2 only.

Hope 5.2 (renormalized energy estimate, any d , any u). *Under ?? 6.3?? 6.4, (8) closes on G_N : $G_N^t \leq G_N^0 + o(N^{s/d-1})$ in Regime P, with no exponential factor and no additive o_N . Consequently $F_N^t = \bar{F}_N^t + o(N^{s/d-1})$: the modulated energy follows the transported self-energy instead of being bounded by a growing multiple of it.*

Two special cases cost nothing beyond [Fact 2.2](#) and are worth extracting first: for dilational $u = \lambda(t)x$ the identity $\text{Comm}_N(a\text{Id}) = -2saF_N$ is exact, so (8) is an explicit ODE plus dissipation; and for $\text{div } u \equiv 0$ (2D Hamiltonian log case) $\bar{F}_N$ is constant, so $G_N = F_N - \text{const}$ and only the shear term remains.

Hope 5.3 (the field theorem's γ). *In $[N]$ the exponent $\gamma = m_* + d + 3$ is dictated by $\|v\|_*$ in the pathwise bound, applied mode by mode. In (6) the test function enters only through ∇v paired against measures, and in (5) through $\|v\|_{L^2(\mu_N)}$ and $\|h_v\|_{C^{\gamma'}}$, with h_{ϕ_k} gaining $|k|^{-1}$ from the Riesz potential. So the commutator step should no longer set γ ; what remains is adjoint regularity and martingale summability.*

Hope 5.4 (Regime P, positive temperature). *Gradient flow, $\beta_{\text{eff}} = \beta N^{s/d-1} \rightarrow 0$, i.i.d. data. The local structure is a Poisson cloud with a hole of size $\ell_B = (\beta/N)^{1/s}$, so S_N° is negligible and e_N concentrates; the commutator is then computed rather than bounded, giving the conclusions of $[N]$ with Assumption MF replaced by $\sigma_N \mathfrak{e}_N \rightarrow 0$, $\mathfrak{e}_N$ the size of the mean commutator. Conjecturally $\mathfrak{e}_N \asymp \beta N^{-1} \vee \beta^{d/s-1} N^{1-d/s}$, hence $s < 2d/3$ for $\sigma_N = \sqrt{N}$, with an explicit Debye correction beyond.*

This is the route I over-sold last time. It is conditional on the static inputs [Missing ingredient 6.6](#), which are weak-coupling expansions with small parameter $N\ell_B^d = \beta_{\text{eff}}^{d/s}$ — not rigidity statements, but not written down for Riesz kernels either.

6. MISSING INGREDIENTS

Missing ingredient 6.1 (localized lower bound, $d = 1$). For $w \geq 0$ smooth symmetric, all X , $\mu \in C^1$ bounded below: $F_N^w(X, \mu) \geq \zeta(s)N^{s-1} \int w(x, \mu)^{1+s} - C\|w\|_{C^1}N^{s-1-\delta}$. This is the weighted version of your screening bound; it is what makes both endpoints of (12) usable.

Missing ingredient 6.2 (less singular remainders). $\int_0^t |\mathcal{R}_N^w| dr = o(N^{s/d-1})$ for the kernels appearing in (10): singularity $|z|^{-s}$ with a weight vanishing on the diagonal, or $|z|^{1-s}$.

Missing ingredient 6.3 (local equilibrium of the energy density). We need $|\mathbb{E}\langle \text{div } u, e_N - \bar{e}_N \rangle| \leq \varepsilon_N \|\nabla u\|_\infty N^{s/d-1}$ with $\varepsilon_N \rightarrow 0$, i.e. mesoscopic averages of the local energy measure concentrate on $\varepsilon_s \mu^{1+s/d}$. This is what (8) needs beyond [Missing ingredient 6.4](#), and it is where the difficulty has moved — correctly, since it is a statement about local structure and not about a sign. It requires localizing e_N to mesoscopic boxes of side ℓ , and the localization error is exactly the almost-additivity surface term of the static free-energy program (the “points cannot be cut” defect), of relative size $\ell^{-1}N^{-1/d}$ per box against a bulk of $N^{s/d-1}$; this fixes the smallest ℓ one can afford and is the one place where the boundary difficulty of the static problem re-enters the dynamics.

Missing ingredient 6.4 (isotropy of the local stress). $|\mathbb{E}\langle E_u^\circ, S_N^\circ \rangle| \leq \varepsilon_N \|\nabla u\|_\infty N^{s/d-1}$ with $\varepsilon_N \rightarrow 0$. This is not a bound to be proved on S_N° directly, because of the following.

Lemma 6.5 (rotation). *Let Σ be a symmetric traceless $d \times d$ matrix. If $R\Sigma R^T = \Sigma$ for a rotation R of order $k \geq 3$ in some plane and, in $d \geq 3$, for the generators of a group acting irreducibly on $\mathbb{R}^d$ (e.g. the cubic group), then $\Sigma = 0$. Consequently, if the conditional law of the local pair structure around a mesoscopic box is invariant under such rotations, then $\mathbb{E}S_N^\circ = 0$ on that box.*

Proof. In the plane of R , Σ restricted is $\begin{pmatrix} a & b \\ b & -a \end{pmatrix}$ (traceless part), and $R_\theta \Sigma R_\theta^T$ rotates (a, b) by angle 2θ ; invariance under $\theta = 2\pi/k$, $k \geq 3$, forces $a = b = 0$. In $d \geq 3$ an invariant symmetric

matrix under an irreducible action is a multiple of the identity (Schur), hence its traceless part vanishes. The pair measure $\iint |z|^{-s-2} (z \otimes z)^\circ \delta_m d\rho_N^{\otimes 2}$ localized to the box is a symmetric traceless matrix which is covariant under rotation of the configuration, so its expectation is invariant and vanishes. $\square$

So the ingredient is *isotropy of the local structure*. In Regime P it is a relaxation statement — the local structure equilibrates on $\tau_{\text{relax}} \asymp \beta N^{-2/d}$ and the strain imposes anisotropy $\asymp \tau_{\text{relax}} \|\nabla u\|_\infty$, so $\varepsilon_N \asymp \beta N^{-2/d}$. In Regime R it is the angular, curvature-weighted local law that was the single hypothesis of the static quadrupole reduction, now required at density μ^t uniformly in t ; nothing in the dynamics makes it easier. It is the dynamic form of the shear obstruction.

Missing ingredient 6.6 (static inputs for [Hope 5.4](#)). For the modulated Gibbs measures

$$Q_\theta \propto \exp \left(-\frac{\beta}{2N} \sum_{i \neq j} \mathbf{g}_\theta(x_i - x_j) + \sum_i (\text{one-body}) \right) \mu^{\otimes N}, \quad \frac{1}{2} \mathbf{g} \leq \mathbf{g}_\theta \leq \frac{3}{2} \mathbf{g} :$$

(i) $\mathfrak{z} := \frac{\log Z_N}{\beta N} = o(N^{s/d-1})$; (ii) $\sup_\theta |\mathbb{E}_{Q_\theta} \text{Comm}_N(v)| = o(N^{s/d-1})$; (iii) sub-Gaussian linear statistics. The singularity works in our favour here ($e^{-\frac{\beta}{N} \mathbf{g}} \leq 1$ suppresses close pairs; the tilt is dominated by $\mathbf{g}$); the difficulty is the renormalization against the one-body term.

7. OBSTRUCTIONS

Obstruction 7.1 (shear). *Fact 4.2*: in $d \geq 2$ with $E_u^\circ \neq 0$ the hierarchy does not close, and [Missing ingredient 6.4](#) is exactly the missing piece. Same object as the static quadrupole Q_0 , with the same conditional reduction available (isotropy via rotation symmetry, or the angular local law); nothing dynamic adds to it. If the static quadrupole is resolved, Regime R follows modulo [Missing ingredient 6.3](#); if not, nothing here passes $\beta_{\text{eff}} \ll 1$ in $d \geq 2$.

Obstruction 7.2 (Coulomb borderline at zero temperature). $\beta = \infty$, $s = d - 2$: microscopic motion is $O(1)$ per unit time, so the local structure is neither transported nor relaxed, and there is no temperature to expand in.

Obstruction 7.3 (non-gradient flows). [Hope 5.4](#) needs an entropy bound relative to Q_N , supplied by the modulated free energy only in the gradient case; and (10) loses the sign of MD_N^w .

8. ORDER OF WORK

- (1) ?? 2.1?? 2.2?? 3.2 into $[N]$ as exact statements, and rerun the energy identity for $\mathbf{G}_N$ rather than $\mathbf{F}_N$. Immediate payoff: no compression forcing, no floor, and [Hope 5.3](#) (γ no longer set by the commutator). No new ingredient required; the dilational and incompressible cases close on the spot.
- (2) [Hope 5.1](#): $d = 1$, $\beta = \infty$, lattice data, via (11)–(12). Needs ?? 6.1?? 6.2. Self-contained and a template for the rest.
- (3) ?? 6.3?? 6.4 in Regime P — the two ingredients of [Hope 5.2](#). For [Missing ingredient 6.4](#) use the transferred rotation argument and prove only relaxation ($\varepsilon_N \asymp \beta N^{-2/d}$); for [Missing ingredient 6.3](#) fix the mesoscale from the additivity-defect count first.
- (4) [Hope 5.4](#) and its static inputs [Missing ingredient 6.6](#), starting from $2s < d$ and small β .
- (5) Leave shear in $d \geq 2$ and the Coulomb borderline stated as open.